When Does Auxiliary Data Help? Global Geometry and Optimal Mixing under Task Mismatch

Anonymous CVPR submission

Paper ID *****

Abstract

When should auxiliary data from another embodiment be used, in what proportion, and how does that decision change with data scale? We study these questions through the global shape of the target risk in a dual-domain full-ridge model with task mismatch. A fixed-dimension expansion of the actual risk and its first two derivatives links population structure to finite-sample optima. At zero mismatch, we classify log-convexity and flat population risks. In general, we prove that one coordinate cannot produce multiple valleys, while two coordinates suffice: a certified example has two local minima, and a one-parameter family exhibits valley creation, annihilation, and two distinct global minimizers. The stationary equations reduce to polynomial root isolation, giving exact global certificates rather than grid-based claims. For finite samples, a nondegenerate interior optimum admits a first-order weight correction, multiple population optima are resolved by the smaller R_1 branch, and matched spectra with zero mismatch yield the exact form $R_1 = A(1-w)^2 + Bw^2$ with explicit optimal weight, risk, and gain. Exact algebra, interval certificates, and conditional Monte Carlo support the results. A unified closed-form classification of all parameters, general finite-sample shape theory, and robot-policy validation remain open.

1. Introduction

Robot demonstrations are expensive, while egocentric human recordings and other auxiliary embodiments are cheap and abundant. Mixed training can exploit this asymmetry, but auxiliary data can also produce negative transfer. The question is therefore not merely whether more data helps, but *when it should be used, in what proportion, and how that decision changes with data scale*.

We study a dual-domain model with Gaussian covariates, a shared spectral basis, task-vector mismatch δ, and domain-specific noise. Standard full-ridge regression is the main estimator; diagonalized ridge is retained only as an interpretable special case. The model is deliberately simple so that the risk geometry and the global optimum can be characterized rather than inferred from a local derivative or a coarse grid. Closed-loop robot policies are outside the scope of this paper.

From risk expansion to global shape. Our first result is a fixed-dimension expansion of the actual full-ridge risk and its first two derivatives (Theorem 1). Instead of then assuming a unique nondegenerate interior minimizer, we analyze the population risk itself. At zero mismatch, log-convexity gives a complete uniqueness and flatness classification. In general, one coordinate cannot produce multiple valleys, whereas two suffice. A certified two-dimensional example and a one-parameter family show local minima separated by a local maximum, valley creation and annihilation, and two distinct global minimizers. These are exact statements about the population risk, not numerical shape conjectures.

Algebraic global certification. The population derivative can be cleared of positive denominators to obtain a polynomial stationary equation of degree at most $3p-2$. For fixed parameters, complete real-root isolation followed by comparison of all stationary and boundary candidates certifies the global optimum without assuming convexity or unimodality. This supplies a direct way to check when the nondegenerate perturbation theorem applies and when a different global-shape analysis is required.

Finite-sample optimal selection. When a unique nondegenerate interior population optimum does exist, we derive a first-order sample-size correction to the finite-sample optimal weight and expansions for optimized risk and gain (Theorem 4). When several population minimizers coexist, the smaller R_1 branch wins at first order. When matched spectra and zero mismatch make R_0 flat, the exact quadratic form $R_1 = A(1-w)^2 + Bw^2$ determines the optimal weight, risk, and gain. Higher-order terms are needed only

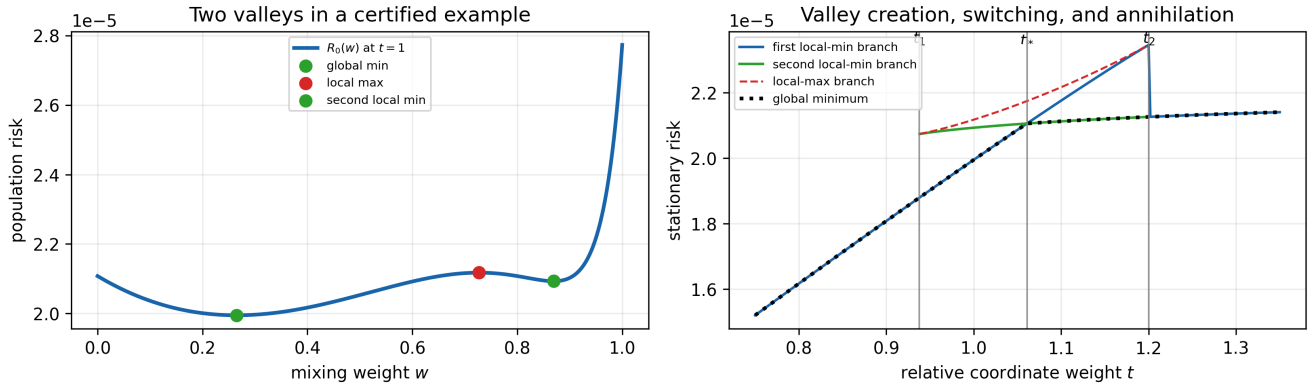


Figure 1. **Global shapes of auxiliary-data mixing risk.** Left: a certified two-dimensional population risk with two local minima and one local maximum; the first minimum is the unique global optimum. Right: stationary branches of a one-parameter family. Valley pairs appear and disappear at t_1, t_2, and at t_* two distinct weights share the global minimum. These are exact algebraic and interval-certified structures, not grid-based shape conjectures.

when these first-order objects also tie or when more precise degenerate rates are requested.

Dosage, exchange, and evidence. At matched spectra, we retain the optimized full-ridge gain expansion and a first-order exchange approximation. An explicitly defined quadratic surrogate admits an exact piecewise dosage rule; this rule is not asserted for the complete full-ridge risk. Validation separates exact algebraic checks, real-root and interval certificates, and conditional Monte Carlo. A unified closed-form classification of all parameters, general R_1/R_n shape theory, the proportional regime, and robot-policy validation remain open.

Contributions. Our contribution is not the first introduction of task mismatch or weighted transfer regression. It is a global analysis of what those risks look like and how their optima are selected: (i) a fixed-dimension full-ridge risk and derivative expansion for dual-domain mixing with task mismatch; (ii) log-convex and flat special cases, a minimum-dimension multimodality result, and polynomial global certification for fixed parameters; (iii) a first-order correction at nondegenerate interior optima, R_1 branch selection at multiple population optima, and an exact flat-R_0 theory at matched spectra and zero mismatch; (iv) matched-spectrum gain and surrogate-dosage formulas, supported by exact algebra, interval certificates, and conditional Monte Carlo. Proofs, exact parameters, and numerical protocols are in the supplementary material.

2. Related work

Egocentric human data for robot learning. Egocentric video has long served as a representation prior: Ego4D [7] supplies large-scale first-person video, and R3M/VIP-style pretraining [15, 16] distills it into visual features for downstream control. A more recent line injects egocentric data *directly into policy training*: EgoMimic [11] uses hand-mounted grippers whose kinematics match the robot, UMI [4] uses hand-held grippers with visual-inertial pose recovery, DexCap [19] uses portable mocap, and EgoBridge [18] studies mixing in simulation with explicit domain gaps. Cross-embodiment robot corpora [17] face a cousin of the same problem across robot morphologies. These works demonstrate practical benefits, substitution ratios [5], and failure modes, and motivate our complementary regression-level analysis of how task mismatch and sample scale affect an explicitly optimized mixing weight.

Domain adaptation and transfer regression. Classical domain-adaptation theory bounds target error by source error plus a discrepancy and a shared-optimal-hypothesis term [2]; subsequent work studies when quantity compensates for quality, the value of target data, and when representation invariance can hurt [3, 9, 23]. More closely, [21] analyze two-task linear regression under covariate and model shifts in the proportional high-dimensional limit. They characterize the bias and variance of hard parameter sharing, identify positive-to-negative transfer transitions, and study source rebalancing. [12] consider weighted joint ridge regression with multiple heterogeneous sources, deriving deterministic risk equivalents that capture covariance shift, model shift, sample sizes, regularization, and privacy noise. Their formulation contains pooled weighted ridge as a limiting case and includes Bayes-optimal tuning under Gaussian priors. These works therefore already provide precise transfer-risk analyses and source-weighting principles under task mismatch. Our contribution is complementary: a fixed-dimension, second-order analysis of the actual full-

ridge risk as a function of a scalar mixing weight, with uniform derivative control and explicit consequences for optimized risk and interior or boundary optima.

Mixture scaling laws and the present scope. [20] study mixed ridge regression with a shared regression function, heterogeneous covariances and noise levels, and derive minimax and ridge scaling laws under aligned power-law spectra. Sim-to-real co-training analyses provide complementary empirical models of multi-source learning [13]. Relative to this line, we allow an explicit deterministic task mismatch δ and focus on fixed-dimension risk and optimum expansions rather than proportional-regime scaling exponents. Our exchange approximation and the piecewise dosage rule are correspondingly restricted: the former is first order, while the latter is exact only for an explicitly defined quadratic surrogate.

Technical tools. Our analysis builds on the ridge-regression literature under power-law spectra [6, 10], with two differences that matter: the robot-domain risk is Σ_r-weighted (only target-geometry errors count), and the sample-level coupling between coordinates induces a *label-feedthrough* term that acts as effective noise (Sec. 3). On the alignment side we use kernel MMD [8, 14] as the alignment-strength knob in experiments; gradient-conflict diagnostics follow the multi-task learning literature [22]; and the impossibility caveats of invariant-representation approaches [1, 23] inform our negative results.

3. A dual-domain model of cross-embodiment mixing

We abstract the problem to its statistical skeleton. Two domains provide training data: the *robot domain* r (target; teleoperated demonstrations) and the *human domain* h (source; egocentric recordings). We write $e \in \{r, h\}$ for the domain index.

Assumption 1 (Data model)**.** Coordinates $j = 1, \dots, p$ index a shared spectral basis. Observations are $x_e = \Sigma_e^{1/2} u_e$ with u_e standard Gaussian and $\Sigma_e = \mathrm{diag}(\lambda_j^e)$ diagonal with power-law spectra

$$\lambda_j^e \asymp j^{-\alpha_e}, \qquad \lambda_1^e = 1, \tag{1}$$

and throughout we abbreviate $a_j = \lambda_j^r$, $b_j = \lambda_j^h$.

Assumption 2 (Labels and mismatch)**.** $y_r = \langle \beta^*, x_r \rangle + \varepsilon_r$ in the robot domain; $y_h = \langle \beta^* + \delta, x_h \rangle + \varepsilon_h$ in the human domain. The *task mismatch* δ absorbs retargeting error, kinematic differences, and residual appearance-induced shifts; its *alignment* with the task is $c = \delta^\top \beta^*$. For each domain e and sample i, $\varepsilon_{e,i}$ has mean zero and variance σ_e^2, is independent of the design and of other samples, and has finite fourth moment. The fourth-moment assumption is used only for Cauchy–Schwarz and complementary-event bounds. The parameter β^* obeys an ellipsoid constraint $\sum_j j^{2s} \beta_j^{*2} \le 1$ with $s > 0$.

Assumption 3 (Mixture estimator)**.** Given n_r robot and n_h human samples and a mixing ratio $\lambda \in [0, 1]$, per-sample weights are $w_r = (1 - \lambda)/n_r$, $w_h = \lambda/n_h$. The main estimator is the *standard full-ridge* solution

$$\hat{\beta} = \left(w_r X_r^\top X_r + w_h X_h^\top X_h + \mu I \right)^{-1} \times \left(w_r X_r^\top y_r + w_h X_h^\top y_h \right), \qquad \mu > 0. \tag{2}$$

The diagonalized estimator, which retains only the diagonal of the sample Gram matrix, is analyzed separately as an interpretable special case. The coefficient μ is the ridge penalty and stands in for explicit or implicit regularization.

The evaluation criterion is the *robot-domain excess risk*, weighted by the robot geometry:

$$\mathcal{R}_T(\lambda) = \mathbb{E} \left\| \hat{\beta}(\lambda) - \beta^* \right\|_{\Sigma_r}^2. \tag{3}$$

Only errors in directions the robot actually visits are penalized — this weighting turns out to be essential for the phase structure in Sec. 4.

Label feedthrough. A subtle point drives the quantitative accuracy of the theory. Although the population covariances are diagonal, the *sample* Gram matrices are not: the label projection T_{ej} picks up contributions from all other coordinates, $T_{ej} = S_{ejj} \beta_{e,j} + \sum_{k \neq j} S_{ejk} \beta_{e,k} + X_{e,:j}^\top \varepsilon_e$. The middle term is mean-zero with variance $n_r \lambda_j^e F_{e,j}$ (and similarly for h), where $F_{e,j} = \sum_{k \neq j} \lambda_k^e \beta_{e,k}^2$ is the domain's signal energy on the coordinates *other than* j ($\beta_h = \beta^* + \delta$). Out-of-coordinate signal is therefore indistinguishable from noise for coordinate j, giving rise to the *effective noise*

$$\tilde{\sigma}_{e,j}^2 = \sigma_e^2 + F_{e,j}. \tag{4}$$

A heavier-tailed human spectrum (small α_h) inflates $F_{h,j}$: broad coverage carries a hidden per-coordinate SNR cost, which competes with its coverage benefit — this tension is the microscopic origin of the phase diagrams in Sec. 4. Where the coordinate dependence is immaterial we abbreviate $\tilde{\sigma}_{e,j}^2$ (resp. $F_{e,j}$) as $\tilde{\sigma}_e^2$ (resp. F_e); all numerical evaluations use the exact per-coordinate form.

What the model keeps and discards. The model retains the three gap components that mixture practice manipulates — spectra (coverage), mismatch δ (embodiment/retargeting), and noise — and discards the rest (nonlinear observation maps, temporal structure, closed-loop

compounding). Sec. 5 therefore validates at three levels: within-model Monte-Carlo, Adam-trained estimators on synthetic data, and stress tests of the assumptions; policy-level evaluation is companion work and is not claimed here. The diagonalized estimator is a controlled approximation of the full-ridge solution in Assumption 3; we quantify its gap (8–28% on the weighted risk) in the supplementary. The full-ridge risk and its first two derivatives are expanded at fixed dimension in Theorem 1. The earlier diagonal-motivated full-ridge surrogate is retained only as a numerical comparator, not as the proved second-order object.

4. Full-ridge risk geometry and optimal data mixing

The main object is the full-ridge target risk $R_n(w)$ and its global minimizers over $w \in [0,1]$. Diagonalized ridge is retained only as an interpretable special case. Proofs and exact algebraic certificates are deferred to the supplementary material.

4.1. Actual full-ridge risk at fixed dimension

Theorem 1 (Second-order full-ridge risk, T1-c)**.** *Consider the standard full-ridge estimator under the Gaussian design and noise model, with fixed p, fixed $\mu > 0$, and fixed $\rho = n_h/n_r$. Then, uniformly for $w \in [0,1]$,*

$$R_n(w) = R_0(w) + \frac{R_1(w)}{n_r} + o(n_r^{-1}), \tag{5}$$

where R_0 and R_1 are explicit matrix coefficients derived in the supplementary. The first two derivatives satisfy the same uniform expansion.

This is an expansion of the actual full-ridge risk, not of a diagonalized object or a numerical surrogate. It connects population geometry to finite-sample optima without requiring a single global shape.

4.2. Global shape of the population risk

With $D_j(w) = (1-w)a_j + wb_j + \mu$, the population risk has the coordinate representation

$$R_0(w) = \sum_j a_j \frac{(wb_j\delta_j - \mu\beta_j^*)^2}{D_j(w)^2}. \tag{6}$$

Proposition 2 (Log-convexity and flatness at zero mismatch)**.** *Suppose $\delta = 0$ and let $\mathcal{J} = \{j : a_j\beta_j^{*2} > 0\}$. Then R_0 is log-convex on $[0,1]$. If some $j \in \mathcal{J}$ has $a_j \neq b_j$, then R_0 is strictly log-convex and has a unique minimizer. If every $j \in \mathcal{J}$ has $a_j = b_j$, or $\mathcal{J}$ is empty, then R_0 is constant and every weight is optimal. In the strictly log-convex case, the signs of $R_0'(0^+)$ and $R_0'(1^-)$ determine whether the unique optimum is at zero, one, or an interior point.*

Proposition 3 (Minimum dimension for multiple valleys)**.** *For $p = 1$, $R_0'(w) = L(w)/D(w)^3$ with affine L, so there is at most one nondegenerate local valley. In contrast, a certified two-dimensional example satisfying Assumptions 1–3 has three simple interior stationary points*

$$w_1 \approx 0.264842, \quad w_2 \approx 0.727011, \quad w_3 \approx 0.869112, \tag{7}$$

which are respectively a local minimum, a local maximum, and a local minimum. Interval risk comparison proves that w_1 is the unique global minimizer. Thus $p = 2$ is the minimum dimension for multiple valleys.

The example does not merely show multiple local minima. Varying its second-coordinate weight t gives valley-pair creation at $t_1 \approx 0.9375848$ and annihilation at $t_2 \approx 1.2003419$. Between these critical values, the global optimum switches branches, and at some $t_* \in [1.06085925, 1.06085926]$ the two local minima have equal risk: two different weights are global minimizers.

Exact global certification at fixed parameters. For each coordinate, $R_{0j}'(w) = L_j(w)/D_j(w)^3$ with affine L_j. After clearing positive denominators, the stationary equation becomes

$$P(w) = \sum_j a_j L_j(w) \prod_{k\neq j} D_k(w)^3 = 0, \qquad \deg P \leq 3p-2. \tag{8}$$

For fixed parameters, isolating all real roots of P in $(0,1)$ and comparing them with $w = 0, 1$ certifies the global minimizers without assuming convexity or unimodality. Degenerate roots and flat cases are handled separately.

4.3. Finite-sample optimal selection

Theorem 4 (Nondegenerate interior optimum)**.** *Suppose R_0 has a unique global minimizer $w_0 \in (0,1)$ with $R_0''(w_0) > 0$. Then, for all sufficiently large n_r, R_n has a unique global minimizer w_n^* on $[0,1]$, with*

$$w_n^* = w_0 - \frac{R_1'(w_0)}{n_r R_0''(w_0)} + o(n_r^{-1}),$$
$$R_n(w_n^*) = R_0(w_0) + R_1(w_0)/n_r + o(n_r^{-1}). \tag{9}$$

Consequently, $\mathcal{G}_n = R_n(0) - R_n(w_n^*)$ satisfies $\mathcal{G}_n = \mathcal{G}_0 + \mathcal{G}_1/n_r + o(n_r^{-1})$, where $\mathcal{G}_0 = R_0(0) - R_0(w_0)$ and $\mathcal{G}_1 = R_1(0) - R_1(w_0)$.

Theorem 5 (R_1 selects among multiple population optima)**.** *Suppose R_0 has two global minimizers $w_1 < w_2$, both with positive local curvature. If $R_1(w_1) < R_1(w_2)$, then for all sufficiently large n_r the global optimum is attained only near w_1, and $w_n^* \to w_1$. The risk gap between the two local branches is $(R_1(w_2) - R_1(w_1))/n_r + o(n_r^{-1})$. If the first-order values tie, R_2 or higher-order analysis is required.*

Theorem 6 (Flat R_0 at matched spectra and zero mismatch)**.** *Let $\Sigma_r = \Sigma_h = \Sigma$ and $\delta = 0$. Then R_0 is constant and there are explicit nonnegative coefficients A, B such that*

$$R_1(w) = A(1-w)^2 + Bw^2. \tag{10}$$

If $A+B > 0$, then R_n is strictly convex for sufficiently large n_r, $w_n^* \to w_1^* = A/(A+B)$, and

$$R_n(w_n^*) = C + \frac{AB}{A+B}\frac{1}{n_r} + o(n_r^{-1}),$$
$$\mathcal{G}_n = \frac{A^2}{A+B}\frac{1}{n_r} + o(n_r^{-1}), \tag{11}$$

where $C = R_0$ and $\mathcal{G}_n = R_n(0) - R_n(w_n^*)$. If $A = B = 0$, the optimal weight and gain require higher-order analysis.

4.4. Full-ridge gain and surrogate dosage

For matched spectra, the full-ridge optimized gain admits $\mathcal{G}_n = G_0 + G_1(\rho)/n_r + o(n_r^{-1})$ at fixed ρ, with $G_1(\rho) = D_0 + D_1/\rho$. A first-order exchange approximation follows by equating target-only and optimized-mixture risks; its pole is a statement about the approximate equation and a fixed-μ baseline, not a uniform actual-exchange guarantee.

An explicitly defined quadratic surrogate

$$\widetilde{R}(w) = R_0(w) + \frac{C_r(1-w)^2}{n_r} + \frac{C_h w^2}{n_h} \tag{12}$$

admits an exact piecewise dosage rule with an interior hyperbolic branch and a $w = 1$ boundary branch. It omits Gram-fluctuation terms that can be of the same order as its retained noise terms, so this identity describes $\widetilde{R}$, not the complete full-ridge risk expansion.

4.5. Mechanism and scope

Alignment, variance reduction, and spectral coverage explain how auxiliary data changes risk, but they do not imply a single-valley landscape. A positive zero-mixing derivative cannot by itself justify discarding the source, and a unique global minimizer does not imply unimodality. The remaining open problems are a unified closed-form classification of all parameters, general R_1/R_n shape theory, higher-order degenerate rates, tuned regularization, the proportional regime, and robot-policy validation.

5. Numerical validation

Validation is organized by evidence type: exact algebraic identities, real-root and interval certificates, and conditional Monte Carlo. The first two certify population objects; the third checks actual finite-sample full-ridge risk without treating formula–grid consistency as independent evidence.

Global-shape certificates. For the two-dimensional multimodal example, exact real-root isolation gives three simple interior stationary points at 0.2648422643, 0.7270105700, and 0.8691124964. Rational interval evaluation proves that the first is the unique global minimum, with risk interval $[1.9948444989, 1.9948445856] \times 10^{-5}$; the second local minimum has a strictly higher interval. The example satisfies exact power-law spectra and the task ellipsoid constraint. Independent NumPy roots differ from the exact roots by at most 1.04×10^{-14}.

For the one-parameter family, the discriminant has two positive roots $t_1 \approx 0.9375848094$ and $t_2 \approx 1.2003419385$. Sturm counts confirm one, three, and one interior roots in the three regions. At $t = 1$ the first branch is globally optimal, while at $t = 1.1$ the third branch wins. Inside $[1.06085925, 1.06085926]$, the branch-risk difference changes sign while all other candidates remain strictly above, proving the existence of two global minimizers. The numerical estimate is $t_* \approx 1.0608592591$.

Full-ridge expansion and optimum checks. The fixed-dimension second-order risk expansion is checked in 22 design-level cells over $p = 1, 2, 20$; all simulated means are within 2.13 Monte Carlo standard errors. Four optimized-risk checks are within 1.59 standard errors. For the two-dimensional aligned example, $w_0 = 0.3708597$, $R_0''(w_0) = 0.13996$, and the predicted scaled weight shift is -2.29329; 200,000 conditional-design replicates at $n_r = 4000$ give -2.29688 ± 0.07803. Actual-E2 paired checks at $n_r = 1000$ give derivatives -0.000435 ± 0.000137 and 0.000469 ± 0.000137 on the two sides of the first-order prediction.

Flat and multiple optima. For matched spectra and zero mismatch, 200 random configurations give maximum R_1 quadratic-identity error 1.82×10^{-12} and maximum flat-R_0 error 3.55×10^{-15}. Four conditional-risk gain checks at $n_r = 1000, 4000$ are within approximately 1.18 standard errors. For the multiple-optimum example, the branch values are $R_1(w_1) = 1.56977$ and $R_1(w_2) = 5.14388$; at $n_r = 1000$, the predicted branch gap is 0.00357412 and 20,000 conditional-risk replicates give $0.00357638 \pm 0.00000078$, confirming the selected branch.

Surrogate dosage and exchange. Across 16 configurations, the closed-form surrogate optimum agrees with grid minimization of the same surrogate to within 1.49×10^{-10} in gain, including a boundary crossing at $n_h = 1779.17$. This checks only the algebraic implementation. A separate full-ridge conditional-risk check gives optimized gain 0.0122262 ± 0.0000066 versus prediction 0.0122186. Exchange checks evaluate the first-order equation, including finite and no-finite-solution examples.

6. Discussion

What the theory buys a practitioner. The main output is not a fixed source fraction or a zero-mixing derivative test. It is a way to distinguish single-valley, multivalley, boundary, multiple-optimum, and flat regimes before choosing a mixing weight. Exact root isolation can certify global candidates for fixed parameters; finite-sample corrections then determine local shifts and select among population-optimal branches. Under matched spectra and fixed regularization, the gain expansion and first-order exchange approximation quantify data value, while the quadratic surrogate provides an interpretable dosage rule whose accuracy requires separate evaluation.

Limitations. The analysis uses fixed dimension, Gaussian covariates, a shared spectral basis, and fixed regularization. The log-convex classification is restricted to zero mismatch; the closed-form branch diagram currently covers a specific two-coordinate family, not the entire parameter space. General R_1/R_n shape theory, higher-order rates at degenerate or multiple optima, tuned regularization, the proportional regime, and robot-policy validation remain open. The surrogate dosage rule omits Gram-fluctuation terms of the same asymptotic order as its retained noise terms and is not a complete full-ridge dosage guarantee.

Outlook. The next priorities are a broader classification of R_1 and R_n, higher-order degenerate perturbation theory, tuned-regularization baselines, and evaluation with independently estimated parameters on native cross-embodiment tasks. Such experiments should report both open-loop risk and closed-loop performance.

Conclusion. We presented a global analysis of full-ridge auxiliary-data mixing. Actual-risk expansions connect population geometry to finite-sample optima. Zero-mismatch log-convexity, minimum-dimension multimodality, exact algebraic certificates, and first-order branch selection replace a single assumed risk shape with a hierarchy of global structures and their consequences. The claims remain limited to the stated model and evidence hierarchy.

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